
Step-Level Gradient Masking for GRPO: Selective Optimization of Reasoning Trajectories

Youhan Lalwani

Department of Artificial Intelligence & Data Science
Thadomal Shahani Engineering College
youhanlalwani@gmail.com

Mahesh Patel

Department of Artificial Intelligence & Data Science
Thadomal Shahani Engineering College
pate12005mahesh@gmail.com

Himani Deshpande

Department of Artificial Intelligence & Data Science
Thadomal Shahani Engineering College
himani.deshpande@thadomal.org

Abstract

Group Relative Policy Optimization (GRPO) has recently emerged as an effective algorithm for Reinforcement Learning from Verifiable Rewards (RLVR), allowing for improvements in logical reasoning, more specifically in mathematical reasoning in large language models (LLMs) without supervised reasoning traces. It does, however, face a credit assignment problem, in that it applies gradient updates uniformly across every token, including trivial formatting steps and already-mastered reasoning prefixes thereby wasting gradient capacity. We propose a step-level gradient masking framework for GRPO that restricts policy updates to structured reasoning steps where the model is still actively uncertain, instead of updating all tokens uniformly. The framework decomposes masking into three components: a step-generation protocol, a masking function, and a routing policy driven by a calibrated threshold. We instantiate the framework with two routing signals: **KL-GRPO** (KL-Divergence GRPO), which routes via per-step KL divergence from a reference model, and **EA-GRPO** (Entropy-Aware GRPO), which routes via the policy’s own Shannon entropy and requires no reference model pass. Training Qwen3-4B on 2,000 stratified RLVR problems with Low-Rank Adaptation (LoRA), EA-GRPO achieves 72.94% on MATH-500 (+1.04% over standard GRPO, paired bootstrap $p = 0.0423$) and 15.83% on AIME 2025 (+2.83%), while KL-GRPO attains the smoothest gradient trajectory and lowest format error rate (3.4%). These results establish step-level gradient masking as a principled, low-overhead design point for GRPO-family training that improves generalization on hard mathematical reasoning tasks.

1 Introduction

The introduction of Reinforcement Learning from Verifiable Rewards (RLVR) has reshaped LLM training and fine-tuning for various tasks including but not limited to code generation, formatting and mathematical Reasoning. Methods such as DeepSeek-R1 [1] and subsequent GRPO-based

systems [2] have shown that verifiable reward signals, in which a symbolic checker confirms whether an answer is correct, can drive dramatic accuracy improvements without human-labeled reasoning traces. At the core of these methods is Group Relative Policy Optimization (GRPO) [2], which samples k completions per input, computes group-relative advantages, and backpropagates gradients uniformly across every token in every trajectory.

This uniform treatment, however, applies the same optimization pressure to tokens with substantially different learning signals. Early steps in mathematical reasoning chains, such as problem restatement, variable definition, and standard formula recall, often exhibit similar behavior across correct and incorrect trajectories and are typically generated with high model confidence. Thus, these steps have limited room for further improvement, while still receiving gradient updates. Moreover, such indiscriminate fine-tuning may also lead to entropy collapse [3], where repeated fine-tuning concentrates a model’s output distribution around narrower, more formulaic responses.

The steps that should be reinforced are those where the model makes genuine decisions, like selecting computation strategies, evaluating competing simplification paths, or exploring a specific chain of reasoning. At these *branching points*, gradient updates carry meaningful information about what reasoning works and what does not.

We implement this intuition through a *step-level gradient masking framework* for GRPO. The framework outputs completions as reasoning steps, applies a masking function over these steps and then uses a routing policy to determine which step receives gradient updates. We present two instantiations of this framework that use reasoning steps as the smallest level of granularity instead of typical token-level approaches.

KL-GRPO (KL-Divergence GRPO) uses per-step KL divergence from the base model as the routing signal. Steps below the threshold are likely already base-model behavior implying suppressing their gradients prevents reinforcement of already-known content. **EA-GRPO** (Entropy-Aware GRPO) instead uses the policy’s own per-step Shannon entropy. Steps where the model is already confident receive no gradient updates, whereas steps with high entropy (low confidence) receive the full gradient update.

Our contributions are:

1. **A step-level gradient masking framework** for GRPO that is made up of three explicit components: step generation, a masking function, and a routing policy.
2. **KL-GRPO** (KL-Divergence GRPO): an instantiation using per-step KL divergence as the routing signal, with a precalibrated empirically determined threshold.
3. **EA-GRPO** (Entropy-Aware GRPO): an instantiation using the policy’s own Shannon entropy as a dynamic routing signal.
4. **Empirical analysis**: a multi-run evaluation across three benchmarks showing that EA-GRPO yields improved accuracy on MATH-500 and AIME 2025, while KL-GRPO provides superior formatting preservation. We also identify an easy-hard trade-off in which step masking improves accuracy on high-difficulty problems at the expense of low-difficulty problems.

2 Related Work

RLVR and GRPO. DeepSeek-R1 [1] has demonstrated that large language models can develop chain-of-thought reasoning through reinforcement learning with verifiable rewards (RLVR), without requiring human feedback. GRPO [2] provides a lightweight policy-optimization framework for this setting by sampling multiple completions for each prompt and computing group-relative advantages from their rewards, avoiding the need for a separately learned value function. Subsequent work has shown that the basic GRPO formulation is susceptible to optimization pathologies and has proposed modifications to its clipping, sampling, normalization, and entropy behavior, including DAPO [3] and Dr. GRPO [4]. These developments motivate a broader question of how the learning signal produced by GRPO should be allocated within a long reasoning trajectory.

Credit assignment in reasoning RL. Outcome-based RLVR presents a fundamental credit-assignment challenge: GRPO derives an advantage from the final outcome of a sampled trajectory, yet a long reasoning trace contains many distinct decisions that may contribute differently to that outcome.

Recent work has therefore explored finer-grained alternatives to trajectory-level credit. Segment Policy Optimization (SPO) [5] introduces segment-level advantage estimation as an intermediate granularity between trajectory-level GRPO and token-level policy optimization, seeking more precise credit assignment without relying on a learned critic. Other approaches similarly exploit structural or outcome-based information to redistribute credit within reasoning trajectories. These works suggest that the granularity at which learning signals are assigned is itself an important design choice for reasoning RL.

Step-level structure in reasoning trajectories. Reasoning trajectories are not homogeneous sequences of decisions: individual steps can differ substantially in difficulty, uncertainty, and their dependence on preceding reasoning. Consequently, a trajectory-level advantage may be incapable of distinguishing between steps which are already reliable and those that need further reinforcement. While credit can be assigned at different granularities, we focus on the reasoning step as a natural intermediate unit between the full trajectory and individual tokens. Step-level masking allows policy updates to be selectively applied to reasoning decisions while retaining the outcome-based learning signal of GRPO, providing finer control than trajectory-level optimization without requiring the granularity of token-level masking.

3 Method

3.1 Preliminaries: GRPO

Let π_θ denote the policy model and π_{ref} the frozen reference model. For a prompt q , GRPO samples k completions $\{o_1, \dots, o_k\}$ and computes group-relative advantages:

$$A_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r}) + \varepsilon} \quad (1)$$

where $\mathbf{r} = \{r_1, \dots, r_k\}$ are scalar rewards. The standard GRPO loss is:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}[\min(r_t A, \text{clip}(r_t, 1-\varepsilon, 1+\varepsilon)A)] + \beta \cdot \mathcal{L}_{\text{KL}} \quad (2)$$

where $r_t = \pi_\theta(a_t)/\pi_{\text{old}}(a_t)$ is the token-level importance ratio. This loss is applied identically to every token in every completion.

3.2 Step-Level Gradient Routing Framework

We propose **Step-Level Gradient Routing (SLGR)**, a general framework for step-level gradient masking in GRPO that replaces the default uniform token weighting with a selective, step-granular masking policy. The framework has three distinct components:

3.2.1 Step Generation

Completions are structured so that the model outputs its chain-of-thought in discrete, programmatically detectable steps. While models can be fine-tuned to output this structure natively, in our instantiations we achieve step boundaries by instructing the model via system prompts to produce reasoning within `<think>` blocks, with each step enclosed in `<step>...</step>` tags, followed by a final response enclosed in `<answer>...</answer>`. LaTeX display equations are represented using `$$...$$` blocks. Step boundaries are detected by applying regular expressions to decoded completion text and mapping character-span offsets to token indices via the tokenizer’s offset mapping. This yields an ordered list of token-index ranges $\{(s_1, e_1), (s_2, e_2), \dots\}$ representing atomic reasoning steps. When no valid steps are detected, the mask defaults to all-ones and standard GRPO applies.

3.2.2 Masking Function

Given detected step boundaries and a routing decision, the masking function constructs a weight mask $M \in \{0, 1\}^T$ over completion tokens. In our instantiations, we implement a *Heaviside step function*: tokens prior to the first active step receive weight zero, while tokens from that step onward receive weight one. The gated loss is:

$$\mathcal{L}_{\text{gated}} = \frac{\sum_t M_t \cdot \mathcal{L}_t}{\sum_t M_t} \quad (3)$$

Alternative masking functions are possible within this framework, including soft gates that assign continuous weights to steps, threshold-based gates with a decay factor, or other continuous weighting functions based on signal magnitude. We select the Heaviside step gate for its simplicity, interpretability, and minimal hyperparameter footprint.

3.2.3 Routing Policy

The routing policy determines which detected steps are activated by evaluating a per-step scalar signal against a threshold parameter τ . In our Heaviside implementation, τ directly controls the cutoff of this thresholding function and is treated as a hyperparameter rather than a learned parameter. We calibrate τ offline using the median of step statistics computed over *divergent evaluation items*, which are problems where the baseline adapter and base model disagree on correctness, so that the threshold reflects the empirical distribution of the selected routing signal. We explore two specific routing signals below.

3.3 Instantiation A: KL-GRPO (KL Divergence Routing)

KL-GRPO uses per-step KL divergence from the reference model as its routing signal. For each step s , it computes:

$$\text{KL}_s = \frac{1}{|s|} \sum_{t \in s} \left[\exp(\log \pi_{\text{ref}}(a_t) - \log \pi_{\theta}(a_t)) - (\log \pi_{\text{ref}}(a_t) - \log \pi_{\theta}(a_t)) - 1 \right] \quad (4)$$

The threshold τ_{KL} is the *median* of all per-step KL values across divergent items. The median is preferred over the mean because the KL distribution is right-skewed.

Routing logic. The active step i^* is the first step where $\text{KL}_{s_{i^*}} > \tau_{\text{KL}}$. All tokens before s_{i^*} are suppressed; all subsequent tokens receive the full gradient. Reference logits are reused from the KL regularization term already computed in the standard GRPO forward pass, adding no extra computation. If no step exceeds τ_{KL} , or if fewer than two steps are detected, or if the advantage is zero, the mask defaults to all-ones (standard GRPO).

3.4 Instantiation B: EA-GRPO (Entropy Routing)

EA-GRPO uses the policy’s own per-step Shannon entropy as the routing signal, requiring no reference model:

$$H_s = \frac{1}{|s|} \sum_{t \in s} \left[- \sum_{v \in \mathcal{V}} P(v \mid \mathbf{x}_t; \theta) \log P(v \mid \mathbf{x}_t; \theta) \right] \quad (5)$$

The threshold τ_H is the *median* of per-step entropies across divergent items, chosen for robustness to the right-skewed entropy distribution. In our experiments, $\tau_H = 0.075$ nats.

Routing logic. The active step i^* is the first step where $H_{s_{i^*}} \geq \tau_H$. Steps where the model is already confident (low entropy) are suppressed; steps where the model is uncertain receive the full update. Entropy is computed directly from the policy logits already present in the forward pass, introducing negligible overhead. Edge cases are handled identically to KL-GRPO.

Comparison of routing signals. KL divergence measures deviation from a static reference and can be increased by stylistic variations such as synonym choice, equation spacing, and step reordering that accumulate KL without reflecting genuine reasoning uncertainty. This may cause KL-GRPO to suppress updates on confident steps that merely phrase things differently from the reference. Shannon entropy is reference-free and measures only the sharpness of the policy’s current output distribution, making it a more direct indicator of where the model is still uncertain.

Table 1: Evaluation Accuracy (Mean % $\pm$ Std Dev across runs)

Dataset	Base Qwen3-4B	Baseline GRPO	KL-GRPO (KL)	EA-GRPO (Entropy)
MATH-500 (25 runs)	68.42 $\pm$ 1.33	71.90 $\pm$ 1.10	71.85 $\pm$ 0.83	72.94 $\pm$ 1.11
AIME 2025 (20 runs)	13.50 $\pm$ 4.77	13.00 $\pm$ 3.57	14.00 $\pm$ 3.99	15.83 $\pm$ 5.50
GSM8K (10 runs)	90.36 $\pm$ 0.40	90.86 $\pm$ 0.38	90.67 $\pm$ 0.48	90.60 $\pm$ 0.43

4 Experimental Setup

4.1 Model and Fine-Tuning

We use QWEN3-4B [6] as the base model, selected for its strong mathematical reasoning baseline and VRAM-efficient inference footprint. All three variants are trained with Low-Rank Adaptation (LoRA) [7] to keep the training tractable on a single GPU. The LoRA configuration targets all linear projection layers (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj) with rank $r = 32$, $\alpha = 64$, dropout 0.05. All experiments were conducted on a single NVIDIA L40S GPU (48 GB VRAM) hosted through RunPod.

4.2 Training Data

Training data consists of 2,000 problems sub-sampled from the **AllenAI RLVR Math** dataset [8], stratified across seven subjects (Algebra, Geometry, Number Theory, Precalculus, Intermediate Algebra, Counting & Probability, Prealgebra) and five difficulty levels to ensure balanced representation. This restricted scale reflects GPU compute budget constraints; we discuss scaling implications in Section 6.2.

4.3 Reward Function

We use a compound scalar reward $r \in [0.0, 1.0]$:

$$\begin{aligned}
 r = & \underbrace{0.85 \cdot \mathbb{I}[\text{correct}]}_{r_{\text{acc}} \text{ (reasoning)}} \\
 & + \underbrace{0.05 \cdot \mathbb{I}[\text{think}] + 0.05 \cdot \mathbb{I}[\text{answer}] + 0.05 \cdot \mathbb{I}[\text{step} \geq 2]}_{r_{\text{fmt}} \text{ (format)}}
 \end{aligned} \tag{6}$$

Answer correctness is verified via a four-tier cascade: exact string matching after LaTeX normalization, numerical float comparison (tolerance $< 10^{-2}$), and SymPy symbolic simplification via both `parse_latex` and `sympify`.

4.4 GRPO Hyperparameters

All three variants share identical GRPO hyperparameters: group size $k = 4$, KL coefficient $\beta = 0.01$, clip epsilon $\varepsilon = 0.2$, learning rate 10^{-5} , maximum new tokens 1,024, generation temperature 0.7, top- p 0.9, gradient clipping norm 1.0, batch size 2 with 2 gradient accumulation steps. Generation is accelerated via vLLM [9] in colocate mode. Training runs for 2,001 steps with checkpoints saved every 200 steps. Table 2 summarizes hyperparameter values.

4.5 Evaluation Protocol

To suppress stochastic variance, each adapter is evaluated in multiple independent runs at temperature 0.7 with `max_tokens` = 2,048 (MATH-500 [10], GSM8K [11]) or 4,096 (AIME). Run counts are: MATH-500: 25 runs (12,500 evaluations per model); GSM8K: 10 runs (13,190 evaluations); AIME 2025: 20 runs (600 evaluations). We report mean accuracy and standard deviation across runs. Evaluation uses the same answer-matching cascade as training.

Table 2: Hyperparameter Configuration

Hyperparameter	Value
Base model	Qwen3-4B
LoRA rank / alpha	32 / 64
Trainable parameters	$\approx 0.3\%$
Group size k	4
KL coefficient β	0.01
Clip ε	0.2
Learning rate	1×10^{-5}
Max new tokens	1,024
Temperature (training)	0.7
Training steps	2,001
KL-GRPO τ_{KL}	0.0009
EA-GRPO τ_H	0.075 nats

5 Results

5.1 Main Accuracy Results

Table 1 summarizes mean accuracy and standard deviation across all stochastic evaluation runs. To further characterize result reliability, Figure 1 reports 95% confidence intervals computed via the t-distribution across all runs per model per benchmark. EA-GRPO is the strongest overall performer on the harder benchmarks, while all three fine-tuned models saturate GSM8K.

On MATH-500, EA-GRPO achieves 72.94% (95% CI: $\pm 0.46\%$), representing a +2.52% absolute gain over the base model and +1.04% over Baseline GRPO. The improvement is even more pronounced on AIME 2025, where EA-GRPO reaches 15.83% (95% CI: $\pm 2.57\%$) versus 13.00% ($\pm 1.67\%$) for Baseline GRPO and 14.00% ($\pm 1.87\%$) for KL-GRPO, a +2.83% gain over the baseline on a 30-problem Olympiad benchmark. On GSM8K, all adapted models cluster within 0.5% of each other, consistent with near-saturation at $\approx 90.6\%$.

5.2 Statistical Significance: Paired Bootstrap Test

To verify that EA-GRPO’s MATH-500 advantage is not attributable to evaluation variance, we conduct a paired bootstrap significance test over the per-problem evaluation logs. Per-problem mean correctness is averaged across all 25 stochastic evaluation runs, yielding paired vectors $\mathbf{a}_{EA}, \mathbf{a}_{Base} \in [0, 1]^{500}$. The test statistic $\delta_{obs} = \bar{\mathbf{a}}_{EA} - \bar{\mathbf{a}}_{Base}$ is compared against $B = 10,000$ bootstrap resamples of 500 problem indices drawn with replacement. The one-sided p -value is the fraction of resamples where $\delta_b \leq 0$; $p = 0.0423 < 0.05$ confirms statistical significance (Table 3).

Table 3: Paired Bootstrap Test: EA-GRPO vs. Baseline GRPO on MATH-500 ($B = 10,000$ resamples, 25 evaluation runs per model).

Metric	Value
Problems N	500
Evaluation runs per model	25
Bootstrap resamples B	10,000
Baseline GRPO mean acc.	71.90%
EA-GRPO mean acc.	72.94%
Observed δ	+1.04%
Bootstrap std. of δ	0.60%
95% Bootstrap CI	$[-0.13\%, +2.25\%]$
p -value (one-sided)	0.0423

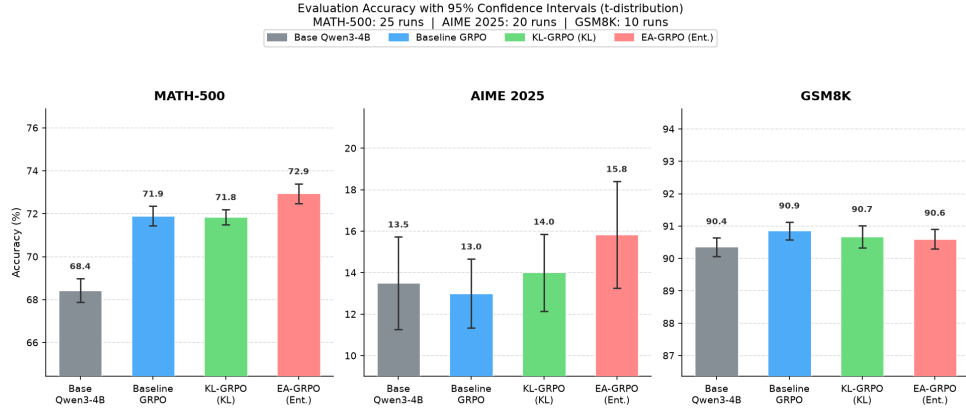


Figure 1: Evaluation accuracy with 95% confidence intervals (t-distribution) across all models and benchmarks.

5.3 Difficulty-Level and Subject Breakdown

Table 4 breaks MATH-500 performance down by difficulty level. On easy problems (Level 1–2), Baseline GRPO outperforms both masked variants; on hard problems (Level 3–5), EA-GRPO is consistently superior.

Table 4: MATH-500 Accuracy by Difficulty Level (%)

Level	N	Base	Baseline	KL-GRPO	EA-GRPO
Level 1	43	86.42	89.58	88.47	88.09
Level 2	90	83.47	87.64	85.42	84.84
Level 3	105	78.78	80.95	81.94	83.50
Level 4	128	67.44	70.84	72.53	73.75
Level 5	134	45.37	49.55	48.84	51.01

EA-GRPO achieves its largest gains at Level 5 (+5.64% over Base, +1.46% over Baseline) and Level 4 (+6.31% over Base). The Baseline GRPO advantage on Level 1–2 reflects a fundamental trade-off: uniform gradient updates across all tokens cause the model to disproportionately optimize for simple, high-frequency patterns, improving performance on easy problems at the cost of hard-problem capacity. Step masking reverses this allocation.

5.4 Training Dynamics and Gradient Stability

Figure 2 shows the per-step L2 gradient norm on a log scale. KL-GRPO’s KL-based masking produces the most stable gradient profile by filtering out low-KL formatting and boilerplate tokens before they contribute to the gradient.

Table 5: Step-Masking Activation Summary (2,001 Training Steps)

Metric	KL-GRPO	EA-GRPO
Routing signal	KL Divergence	Shannon Entropy
Threshold (τ)	0.0009	0.0750
Mean signal	0.0034	0.0683
Avg steps / batch	9.8	10.5
% masked (all steps)	2.2%	1.6%
% masked (non-zero adv.)	7.3%	5.6%
Training steps	2,001	2,001

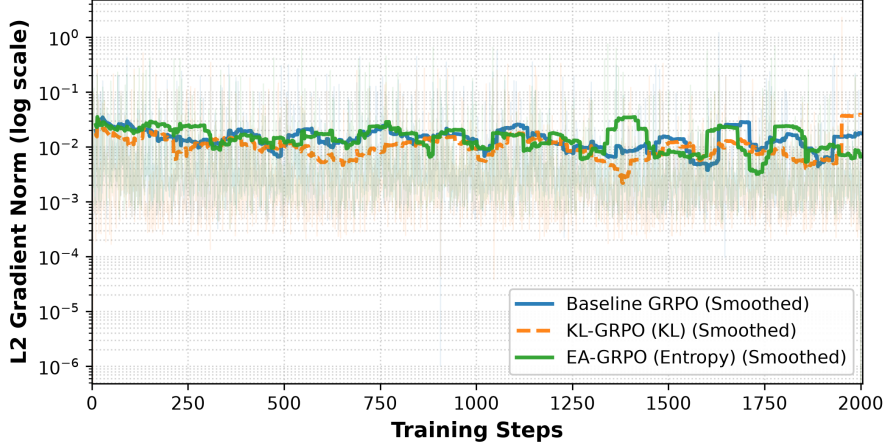


Figure 2: LoRA L2 gradient norm over training (log scale, rolling mean).

6 Discussion and Limitations

6.1 Mechanistic Comparison: KL vs. Entropy Signal

KL divergence measures deviation from a static reference model. Trivial stylistic variations accumulate KL divergence without reflecting genuine reasoning uncertainty, causing KL-GRPO’s mask to open on confident steps that merely phrase things differently from the reference. Shannon entropy avoids this problem by measuring the sharpness of the policy’s current output distribution directly, routing updates strictly to uncertain branching points.

6.2 Limitations and Future Work

We identify the following key areas for future investigation:

1. **Token vs. Step Granularity:** We do not explicitly compare token-level masking to step-level masking in these experiments, leaving the precise impact of semantic granularity undetermined.
2. **Threshold Sensitivity:** Further analysis is needed to determine the sensitivity of model accuracy and gradient stability with respect to the routing threshold τ .
3. **Model & Dataset Scale:** All experiments are conducted on Qwen3-4B trained on 2,000 RLVR samples due to compute constraints. Validation across larger model scales and larger training sets is required to fully assess scalability.

7 Conclusion

We presented a step-level gradient masking framework for GRPO-based reinforcement learning, demonstrating two instantiations: KL-GRPO and EA-GRPO. EA-GRPO achieves superior performance on MATH-500 (72.94%) and AIME 2025 (15.83%) compared to baseline GRPO and KL-GRPO, while maintaining low computational overhead. Future work includes adaptive thresholding and scaling to larger foundation models.

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A Additional Experimental Results

A.1 Evaluation Accuracy Distributions (Boxplots)

Figure 3 shows evaluation accuracy distributions across all stochastic runs for MATH-500, AIME 2025, and GSM8K.

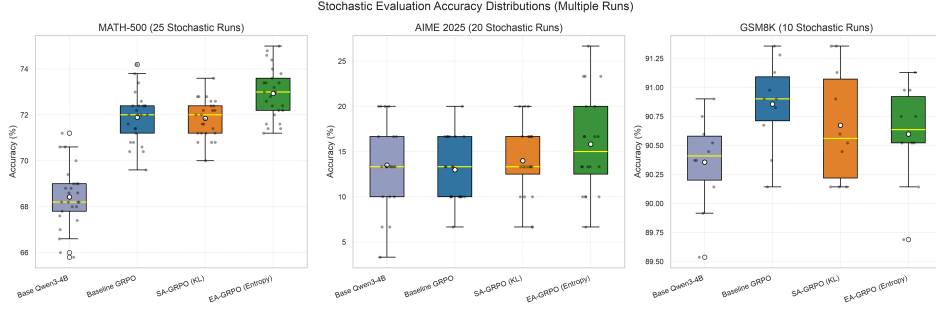


Figure 3: Evaluation accuracy distributions across stochastic runs (MATH-500: 25 runs, AIME 2025: 20 runs, GSM8K: 10 runs). EA-GRPO achieves the highest median and mean on MATH-500 and AIME 2025. GSM8K is effectively saturated at $\approx 90.6\%$.

A.2 Problem-Wise Granular Delta Analysis

To investigate whether accuracy gains stem from uniform incremental progress or targeted problem-solving improvements, Figure 4 shows the per-problem accuracy delta between EA-GRPO and Baseline GRPO across all 500 MATH-500 items, sorted ascending by difficulty level (Level 1 at top to Level 5 at bottom).

Out of 500 problems, EA-GRPO outperforms Baseline GRPO on 106 problems, Baseline GRPO leads on 99 problems, and both models achieve identical accuracy on 295 problems. Crucially, the distribution of positive deltas shifts dramatically toward higher difficulty levels: in Level 4 and Level 5, EA-GRPO dominates the rightward deltas, unlocking difficult multi-step problems that Baseline GRPO consistently fails.

A.3 Subject-Level Accuracy Breakdown

Table 6 details performance across the seven Hendrycks MATH subjects. EA-GRPO leads in four of seven subjects, with its largest margins in Number Theory (+5.03% over Baseline) and Precalculus (+3.29% over Baseline), domains that require precise multi-step logical chains. KL-GRPO holds the top position in Geometry (55.51%), suggesting that KL-anchored masking better preserves spatial and coordinate reasoning chains.

Table 6: MATH-500 Accuracy by Subject (%)

Subject	N	Base	Baseline	KL-GRPO	EA-GRPO
Algebra	124	86.13	88.32	88.19	89.29
Intermediate Algebra	97	53.65	60.54	59.92	59.34
Prealgebra	82	70.34	74.20	72.05	73.80
Number Theory	62	80.39	81.68	85.03	86.71
Precalculus	56	52.64	55.93	56.14	59.21
Geometry	41	51.12	53.66	55.51	55.41
Counting & Prob.	38	66.63	69.58	67.79	69.05

A.4 Training Dynamics and Reward Breakdown

Figure 5 plots cumulative training accuracy over 2,001 steps. Figure 6 breaks down reward components over training.

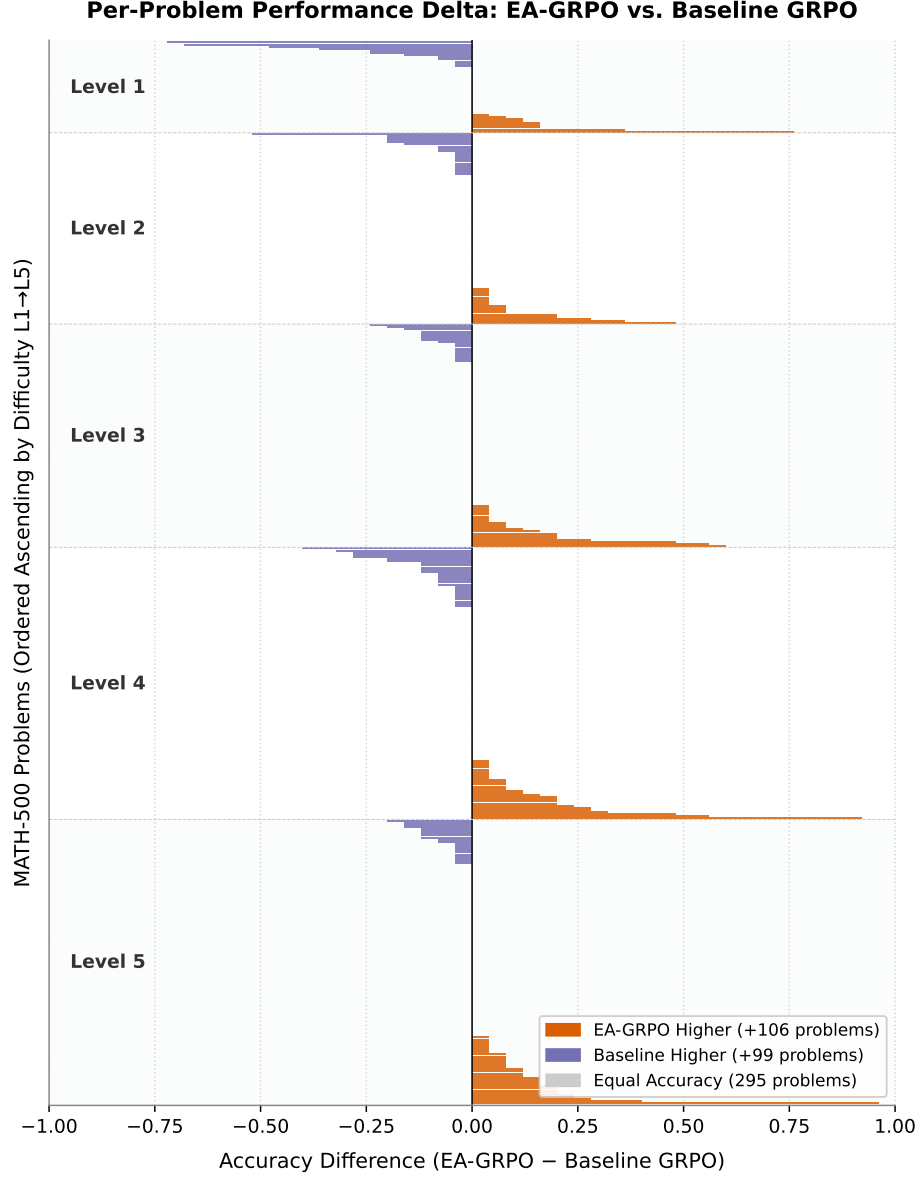


Figure 4: Per-problem accuracy difference (EA-GRPO vs. Baseline GRPO) on MATH-500 across 25 evaluation runs, ordered ascending by difficulty level (L1 to L5).

Table 7: Training Dynamics Summary

Metric	Baseline	KL-GRPO	EA-GRPO
Final train acc. (%)	65.40	64.65	63.13
Mean L2 grad norm	0.0145	0.0112	0.0154
Avg gen. length (tokens)	517.8	515.7	527.2



Figure 5: Cumulative training accuracy (running mean over all completions) over 2,001 steps.

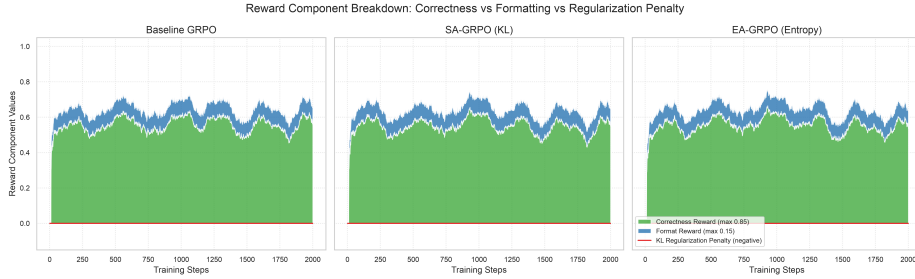


Figure 6: Reward component breakdown over training.

A.5 Error Categorization and Newly Solved Problems

We categorize incorrect responses into three types: truncated (hit token limit), format failures (missing `<answer>` tag), and genuinely wrong (had tag but wrong answer). KL-GRPO achieves the lowest format error rate (3.4% of wrong responses), compared to 4.6% for EA-GRPO and 5.5% for Baseline GRPO.

EA-GRPO uniquely solves 6 problems (success rate $\geq 70\%$ of runs) that the base model fails consistently ($\leq 20\%$ of runs), versus 4 for KL-GRPO and 1 for Baseline GRPO. These newly unlocked problems are predominantly Level 4–5 Algebra and Number Theory problems requiring multi-step symbolic manipulation.

A.6 Computational Cost and Masking Activation

Figure 7 compares response token lengths and training step wall times. Figure 8 details masking gate activation rates. Table 8 summarizes step wall times and format error rates.

Table 8: Mean Step Wall Time and Format Error Rate Summary

Variant	Wall Time / Step (s)	Format Error Rate (%)
Baseline GRPO	10.0	5.5%
KL-GRPO	10.3	3.4%
EA-GRPO	10.1	4.6%

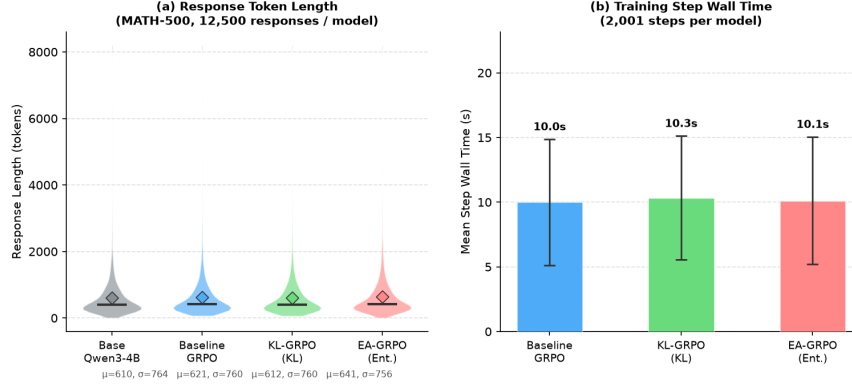


Figure 7: (a) Response token length distributions across 12,500 MATH-500 eval responses per model. (b) Mean training step wall time for fine-tuned models over 2,001 steps.

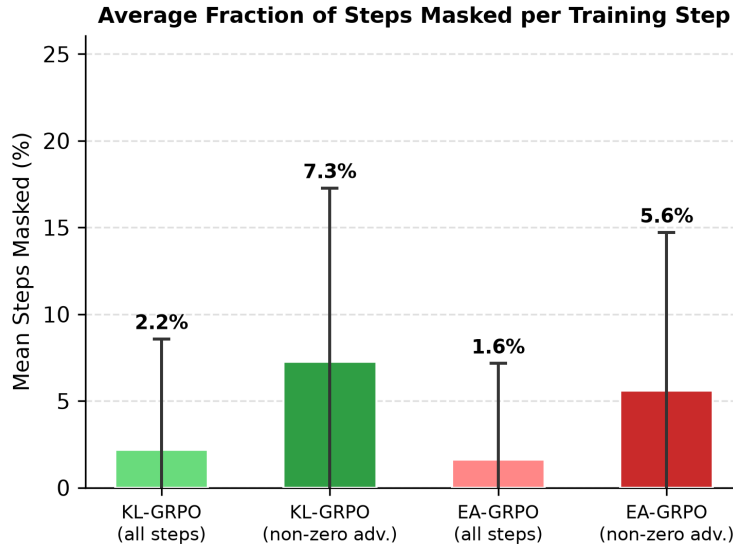


Figure 8: Average fraction of steps masked per training step for KL-GRPO and EA-GRPO, shown for all training steps and restricted to non-zero advantage steps. Error bars show ± 1 std dev.

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